Euclid's Elements — Book I
Proposition 34

Formal Reconstruction — Technical Documentation

Jurek Różański
Military University of Technology, Warsaw, Poland
`jerzy.rozanski@wat.edu.pl`

CGJteamLab

Basic properties of a parallelogram

1 Introduction

Euclid's Proposition I.34 collects the fundamental metric and angular properties of a parallelogram. For a parallelogram $ABCD$, it asserts that opposite sides are equal, opposite angles are equal, and a diagonal bisects the parallelogram.

The current CGJteamLab reconstruction separates these assertions according to the language actually present in the Hilbert development. Opposite sides and opposite angles are packaged directly by segment and angle congruence. The classical statement about the diagonal is represented separately by congruence of the two triangles cut off by that diagonal. No numerical area function is introduced in Proposition I.34, and no `HilbertScissors` term occurs in the production file.

This distinction is important for the later area propositions. At I.34 the project uses triangle congruence as a sufficient synthetic representative of the diagonal bisection statement; beginning with I.35 the documentation must also distinguish geometric congruence from the more general scissors/content calculus.

2 The Classical Configuration

Let $ABCD$ be a parallelogram, with

$$AB \parallel CD, \qquad BC \parallel DA.$$

Draw the diagonal AC. It separates the parallelogram into the triangles ABC and CDA.

The required metric and angular conclusions are

$$AB \cong CD, \qquad BC \cong DA,$$

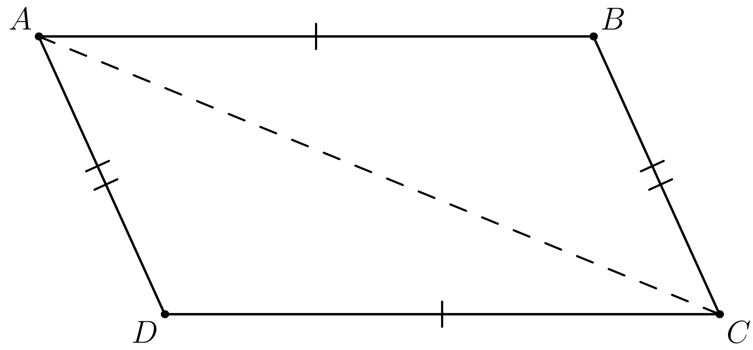

Figure 1: The configuration for Proposition I.34. The single and double chevrons mark the two *parallel* pairs $AB \parallel CD$ and $BC \parallel DA$; they do not assert the opposite-side congruences that I.34 is meant to prove. The dashed segment AC is the diagonal used in both the classical and formal arguments.

$$\angle DAB \cong \angle BCD, \qquad \angle ABC \cong \angle CDA.$$

In the present formalization the diagonal assertion is represented by

$$\triangle ABC \cong \triangle CDA.$$

3 Classical Proof

Euclid's original lettering is different: he writes the parallelogrammic area as $ACDB$ and uses BC as its diameter. After relabelling to the present $ABCD$ picture with diagonal AC, the proof has four logically distinct steps.

First, Proposition I.29 is applied to the two pairs of parallels. With AC as transversal it gives

$$\angle BAC \cong \angle ACD, \qquad \angle BCA \cong \angle CAD.$$

Second, Proposition I.26 is applied to the two diagonal triangles. The two angle equalities and the common side AC give

$$AB \cong CD, \qquad BC \cong DA, \qquad \angle ABC \cong \angle CDA.$$

Third, Euclid does not obtain the other pair of whole opposite angles by a single application of I.26. He combines the two alternate-angle equalities by Common Notion 2: the two parts of $\angle DAB$ are respectively equal to the two parts of $\angle BCD$, hence

$$\angle DAB \cong \angle BCD.$$

Finally, the statement that the diameter bisects the parallelogram is proved as an equality of rectilinear figures by Proposition I.4. In the present lettering one may use $BC \cong DA$, the common side AC, and the included angle equality $\angle BCA \cong \angle CAD$. Euclid therefore obtains the equality of the two triangles on the two sides of the diagonal.

A compact classical DAG is

$$\begin{array}{ccc} & \mathrm{I.29} + \mathrm{I.29} & \\ & \Downarrow & \\ & \text{two alternate-angle equalities} & \\ & \Downarrow & \\ & \mathrm{I.26} & \\ & \Downarrow & \\ & \text{opposite sides + one opposite-angle pair} & \\ \swarrow & & \searrow \\ \text{C.N.2} & & \text{I.4} \\ \downarrow & & \downarrow \\ \text{second opposite-angle pair} & & \text{diagonal bisection.} \end{array}$$

This corrects the earlier compressed description. I.26 supplies corresponding sides and the remaining angle; Euclid's final "bisects the areas" clause is an additional magnitude statement and is closed explicitly by I.4.

4 The Main Formalization Issue: Reusing Parallelogram Theory

The production file `Proposition34.lean` does not replay the classical I.29–I.26 route. By the time Proposition I.34 is reached, the developed `HilbertInterface` already contains a Euclidean theory of parallelograms. In particular, the public theorem reuses

```
ParallelogramOppositeSidesCongruent
ParallelogramOppositeAnglesCongruent
```

and is therefore a thin wrapper.

4.1 Where the Euclidean work actually lives

The important hidden fact in the formal DAG is that `ParallelogramOppositeSidesCongruent` is not a primitive congruence rule. Its core is the reusable theorem

```
parallelogram_second_pair_congruent
```

which proves $BC \cong DA$ from `IsParallelogram Geo A B C D`. The proof constructs a genuine auxiliary point X by segment construction: X lies on the ray from D through A and

$$DX \cong BC.$$

Collinearity transports $AD \parallel BC$ to $XD \parallel BC$. The proof then builds the side-orientation data required by `OnePairParallelCongruentCriterion`; this makes $XBCD$ a parallelogram and hence gives

$$XB \parallel CD.$$

Now both XB and AB are parallels to CD through B. Hilbert Group IV, through

`HilbertEuclideanPlane.parallel_unique`

identifies their carriers. Since X also lies on the carrier DA, incidence uniqueness at the intersection of the carriers AB and DA forces

$$X = A.$$

Substitution gives $BC \cong DA$.

The other side pair is obtained by applying the same theorem to the cyclically relabelled parallelogram $BCDA$, followed by symmetry of segment congruence. This is the precise point at which the Euclidean parallel axiom is consumed in the reusable side theorem.

5 Proof Architecture of the Reconstruction

The actual reusable side-congruence DAG is

$$\begin{gathered} AB \parallel CD, \quad BC \parallel DA \\ \Downarrow \\ \texttt{IsParallelogram}\ ABCD \\ \Downarrow \\ \texttt{parallelogram\_second\_pair\_congruent} \\ \text{segment construction of } X + \text{orientation} + \text{Hilbert IV} \\ \Downarrow \\ \texttt{ParallelogramOppositeSidesCongruent} \\ \Downarrow \\ AB \cong CD, \quad BC \cong DA. \end{gathered}$$

For one pair of opposite angles the interface then uses the diagonal:

$$\begin{gathered} AB \cong CD, \quad BC \cong DA, \quad AC \cong CA \\ \Downarrow \quad \text{HilbertSSS} \\ \triangle ABC \cong \triangle CDA \\ \Downarrow \\ \angle ABC \cong \angle CDA. \end{gathered}$$

The theorem `ParallelogramOppositeAnglesCongruent` applies the one-pair result to $ABCD$ and to the cyclic relabelling $BCDA$, then uses symmetry of angle congruence to package both opposite-angle conclusions.

No geometric case split occurs in `Proposition34.lean`. The only contradiction branch in the diagonal theorem proves non-collinearity, while the hidden $X = A$ argument belongs to the reusable interface theorem above.

6 Lean Formalization

The production module imports only

```
import CGJteamLab.HilbertInterface
```

and exposes two public theorems. There are no proposition-local helper theorems in the file.

The main theorem is deliberately thin:

```
theorem euclid_proposition_34
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D : Geo.Point)
    (hParallelogram : IsParallelogram Geo A B C D) :
    And
      (OppositeSidesCongruent Geo A B C D)
      (OppositeAnglesCongruent Geo A B C D)
```

Its two branches are discharged by

```
ParallelogramOppositeSidesCongruent
ParallelogramOppositeAnglesCongruent
```

The third classical assertion is represented separately:

```
theorem euclid_proposition_34_diagonal
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D : Geo.Point)
    (hParallelogram : IsParallelogram Geo A B C D) :
    TriangleCongruenceResult Geo A B C C D A := by
```

The diagonal proof first derives $\neg\,\mathrm{Collinear}(A, B, C)$ from the disjointness of the parallel point-lines AB and CD. It then takes the two opposite-side congruences, uses reflexivity plus endpoint reversal for the common diagonal, and applies SSS:

```
  have hSides :
      OppositeSidesCongruent Geo A B C D :=
    ParallelogramOppositeSidesCongruent
      Geo A B C D hParallelogram

  have hAC_CA :
      Geo.Congruent A C C A :=
    CongruentSwapSecond
      Geo A C A C
      (hilbert_congruent_reflexive Geo A C)

  have hSSS :=
    HilbertSSS
      Geo
      A B C
      C D A
      hABC
      hSides.1
      hSides.2
      hAC_CA

  exact hSSS.2
```

No area object occurs in either theorem. The diagonal theorem proves a triangle-congruence result; this is sufficient for this particular bisection statement but is not a general formal theory of equal content.

7 Relationship with Earlier Propositions

Classically, I.34 directly uses I.29 twice and I.26, then Common Notion 2 and I.4. The production I.34 file calls none of those proposition names directly. The Euclidean work has instead been moved into reusable Hilbert-level parallelogram theorems.

The relation with I.33 remains close. I.33 is a recognition theorem: from one pair of opposite sides that are parallel and congruent, together with the required orientation data, it obtains a parallelogram. The lower-level side proof used by I.34 invokes the reusable analogue `OnePairParallelCongruentCriterion` for the auxiliary quadrilateral $XBCD$.

8 Hilbert and Book Zero Components Used

At proposition level the visible dependencies are compact:

- `IsParallelogram`;
- `ParallelogramOppositeSidesCongruent`;
- `ParallelogramOppositeAnglesCongruent`;
- `HilbertSSS`;
- incidence uniqueness and extensional point-lines, used in the diagonal theorem to prove A, B, C non-collinear;
- reflexivity and endpoint reversal of segment congruence.

The side-congruence theorem in `HilbertInterface` additionally uses:

- Hilbert segment construction, producing the auxiliary point X;
- `SameRay` and collinearity transport along DA;
- `SameSide` data for the one-pair parallelogram criterion;
- `OnePairParallelCongruentCriterion`;
- uniqueness of a Euclidean parallel through a point;
- uniqueness of an incidence line through two distinct points.

No new Book Zero theorem is declared in `Proposition34.lean`.

9 Formalization Notes

9.1 Axiomatic status

Both public theorems assume

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

The Euclidean strength is used lower in the actual DAG when the two parallels through B to the carrier CD are identified. This is the Hilbert IV step inside `parallelogram_second_pair_congruent`.

The current source contains no proposition-local `axiom` declaration and no `sorry`. This diagrammatic audit did *not* rerun `lake build` or `#print axioms`; historical build statements in the preserved Book1R PDF are therefore not presented here as fresh audit results.

9.2 Hidden configuration data

There are two different levels of hidden data.

At proposition level, `euclid_proposition_34_diagonal` must prove explicitly that A, B, C are not collinear before SSS can be applied. If C lay on AB, it would lie on both disjoint parallel carriers AB and CD.

At reusable-interface level, the stronger hidden witness is the point X constructed on the ray DA. The proof uses genuine `SameRay` and `SameSide` geometry to turn $XBCD$ into an auxiliary parallelogram and then proves $X = A$ by parallel and incidence uniqueness. These side/order facts do not appear in the public I.34 theorem because the interface theorem has already discharged them.

9.3 Geometry, congruence, content, and representation

The layers should not be conflated:

- **Incidence/order/parallel geometry:** the parallelogram carriers, the diagonal AC, and lower in the reusable DAG the point X with its ray and side data.
- **Congruence:** the opposite-side congruences, opposite-angle congruences, and the SSS comparison of the diagonal triangles.
- **Content:** no general content object or scissors-equivalence relation occurs in Proposition I.34.
- **Representation:** the classical bisection clause is represented here by congruence of the two diagonal triangles. This is stronger than merely saying that they have equal content, but it is not the later `HilbertScissorsEquicomplementable` language.

9.4 Reusable interface helpers

The current I.34 DAG uses

```
parallelogram_second_pair_congruent
ParallelogramOppositeSidesCongruent
ParallelogramOppositeAngleCongruent
ParallelogramOppositeAnglesCongruent
```

The first contains the auxiliary construction and Hilbert-IV uniqueness step. The second packages both side pairs by cyclic relabelling. The third uses SSS across the diagonal for one opposite-angle pair. The fourth packages both angle pairs by cyclic relabelling and symmetry.

9.5 Diagrammatic audit

The preserved I.34 figure had the correct quadrilateral and diagonal but used one and two transverse tick marks on opposite sides. Those marks visually asserted

$$AB \cong CD, \qquad BC \cong DA,$$

which are conclusions of Proposition I.34 rather than hypotheses. The audited figure therefore replaces them by conventional single and double *parallel* chevrons. The input geometry now records exactly $AB \parallel CD$ and $BC \parallel DA$.

No second figure is introduced for the hidden point X. The reusable proof constructs X and subsequently proves $X = A$. A panel that drew X as a separate final vertex while simultaneously displaying all the parallel data would depict a configuration excluded by the uniqueness theorem being used; drawing X coincident with A would add no new geometric state. This is therefore proof-state bookkeeping, not an additional proposition-level figure.

Likewise no figure is added for diagonal reflexivity, SSS, cyclic relabelling, segment reversal, or angle-congruence symmetry. These steps change the logical interpretation of the same configuration but do not create a new geometric state. The final audited figure count is one.

10 Dependency Summary

The classical dependency is

$$\begin{array}{c}
\text{I.29} + \text{I.29} \\
\Downarrow \\
\text{I.26} \\
\Downarrow \\
\text{opposite sides} + \text{one opposite-angle pair} \\
\begin{array}{cc}
\swarrow & \searrow \\
\text{C.N.2} & \text{I.4} \\
\downarrow & \downarrow \\
\text{second opposite-angle pair} & \text{diagonal bisection.}
\end{array}
\end{array}$$

The actual production DAG is

$$\begin{array}{c}
\text{Hilbert incidence/congruence/order} + \text{Euclidean parallelism} \\
\Downarrow \\
\texttt{segment construction / SameRay / SameSide} \\
+\ \texttt{OnePairParallelCongruentCriterion} \\
+ \text{ parallel and incidence uniqueness} \\
\Downarrow \\
\texttt{parallelogram_second_pair_congruent} \\
\Downarrow \\
\texttt{ParallelogramOppositeSidesCongruent} \\
\Downarrow \\
\texttt{HilbertSSS} \\
\Downarrow \\
\texttt{ParallelogramOppositeAngleCongruent} \\
\Downarrow \\
\texttt{ParallelogramOppositeAnglesCongruent} \\
\Downarrow \\
\texttt{euclid_proposition_34}.
\end{array}$$

The separate diagonal endpoint is

$$\texttt{ParallelogramOppositeSidesCongruent} + AC \cong CA + \neg\,\mathrm{Collinear}(A, B, C)$$
$$\Downarrow$$
$$\mathrm{HilbertSSS}$$
$$\Downarrow$$
$$\texttt{euclid\_proposition\_34\_diagonal}.$$

Production import:	`CGJteamLab.HilbertInterface`
Public theorems in `Proposition34.lean`:	two
Proposition-local helper theorem:	no
Proposition-local `axiom` declaration:	no
`sorry` in `Proposition34.lean`:	no
New Book Zero declaration in this file:	no
Geometric case split in this file:	no
Hidden constructed point in reusable DAG:	yes, X
Hilbert Group IV used in reusable DAG:	yes
Numerical area semantics:	no
Scissors/content term in I.34:	no
`lake build` rerun for this audit:	no
`#print axioms` rerun for this audit:	no

11 Conclusion

Proposition I.34 is short at the proposition-file level because the developed Euclidean theory of parallelograms already contains its main metric work. The public theorem is a thin wrapper, but the actual dependency graph is not trivial: opposite-side congruence ultimately passes through an auxiliary segment construction, orientation data, a one-pair parallelogram criterion, Hilbert-IV parallel uniqueness, and incidence uniqueness.

Euclid's historical DAG is different. He obtains alternate angles by I.29, uses I.26 for corresponding sides and one opposite-angle pair, Common Notion 2 for the other whole opposite angle, and I.4 for equality of the two diagonal triangles as rectilinear figures. The current Lean development instead reuses the established side theorem and applies SSS across the same diagonal.

The diagrammatic audit leaves exactly one figure. Its role is now precise: it records only the parallelogram hypotheses and the auxiliary diagonal. It no longer displays the theorem's opposite-side conclusions as if they were input data.